RAG Real-Life Use Case

Internal Policy Bot

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Tesla Internal Policy Bot

An AI assistant that answers ethics and compliance questions from approved policy documents, with sources shown.

BUSINESS BENEFITS

1 **Faster answers**

Seconds, not a search through manuals

2 **Easy to Interpret**

Same policy text behind every answer

3 **Auditable by design**

Every answer links to its source

Scattered Documents, Difficult-to-Find Answers

KEY CHALLENGES

1. **Scattered documents**
   Ethics, privacy, security and confidentiality rules sit in separate manuals.

2. **Hard to search**
   Keyword search misses answers when staff use different wording.

3. **Inconsistent reading**
   Different teams interpret the same rule differently.

4. **Unverifiable AI answers**
   Generic chatbots answer fluently without citing company policy.

BUSINESS IMPACT

Lost productive time
Skilled staff searching instead of building.

Compliance exposure
A misread rule can become a violation.

Repeat questions
Ethics teams re-answer routine queries.

Shadow AI risk
Staff turn to unvetted external tools.

Ask in plain English and get a cited answers.

1

Ask naturally

No legal titles or section numbers needed.

2

Search by meaning

Finds the most relevant policy passages.

3

Answer from evidence only

Model must use retrieved text and decline otherwise.

4

Show the sources

Document ID and excerpts shown beside the answer.

REAL DEMO OUTPUT

SYNTHETIC POLICY

EMPLOYEE QUESTION

What is the statutory SLA for processing customer vehicle telemetry deletion requests?

BOT ANSWER

Timeframe: process, execute and verify a complete deletion request within 30 business days of receipt.

Scope: all stored vehicle telemetry logs and associated cloud metadata (GDPR, CCPA, CPRA).

(Cited: TSLA-ETH-2026-02, Section 5.2, 30-Day Statutory Deletion SLA)

SOURCE SHOWN IN THE UI

TSLA-ETH-2026-02 (customer_data_privacy.txt) | Section: Full Document

Four user groups, one simple workflow

Employees

Get policy guidance in seconds

Metric: lookup time, satisfaction

HR & Ethics teams

Fewer repetitive questions

Metric: inquiry ticket volume

Compliance teams

More consistent awareness

Metric: audit vs. expert answers

People managers

Confident team decisions

Metric: manager feedback

WORKFLOW

1 Sign in	2 Ask	3 Retrieve	4 Answer	5 Learn More
Mock login in demo	Type a question in plain English	Top 3 matching policy chunks	Grounded, cited response	Open the source excerpts

Implemented: Streamlit chat UI, source badges, chunk inspector.

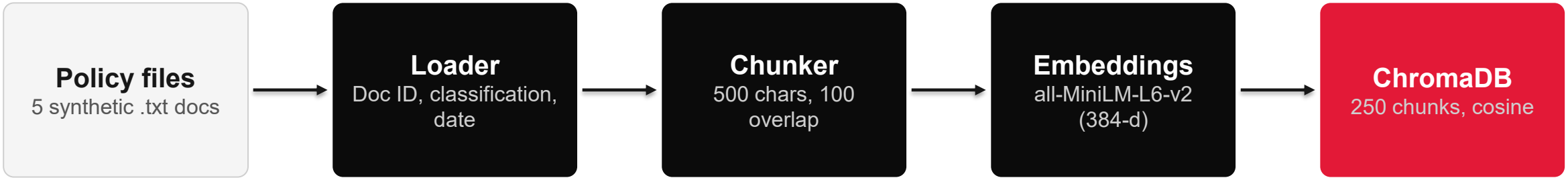
RAG keeps answers grounded and policies easy to update

Approach	Plain-language questions	Grounded in company policy	Shows sources	Updates without retraining
Manual search	No	Yes	Yes	Yes
Keyword search	No	Yes	Yes	Yes
General LLM only	Yes	No	No	No
Fine-tuned LLM	Yes	Partly	No	No
RAG assistant	**Yes**	**Yes**	**Yes**	**Yes**

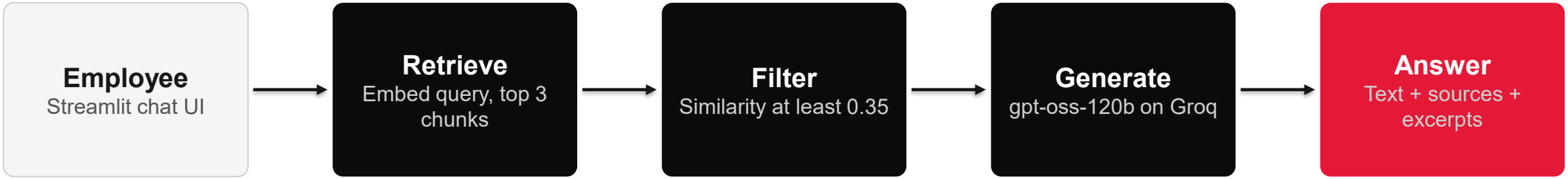
Trade-off: RAG quality depends on retrieval and document coverage. Qualitative assessment.

End-to-End RAG architecture

A. OFFLINE: DOCUMENT PREPARATION



B. REAL TIME: QUESTION ANSWERING



Guardrails in the prompt: answer only from retrieved context, no invented policies, cite document and section, say so when information is missing. **Temperature 0.0 by default.**

LLM VS SLM

SMALL MODEL (SLM)

e.g. 3B-8B

- Lower resource requirements
- Potentially lower latency and cost
- Suitable for simpler, focused tasks
- May require more optimization for complex tasks

LARGE MODEL (LLM)

e.g. 70B-120B

- Handles complex questions
- Better suited to multi-part responses
- Flexible natural-language generation
- Typically higher compute cost

MODELS USED

Embedding

all-MiniLM-L6-v2

384-d vectors

Generation

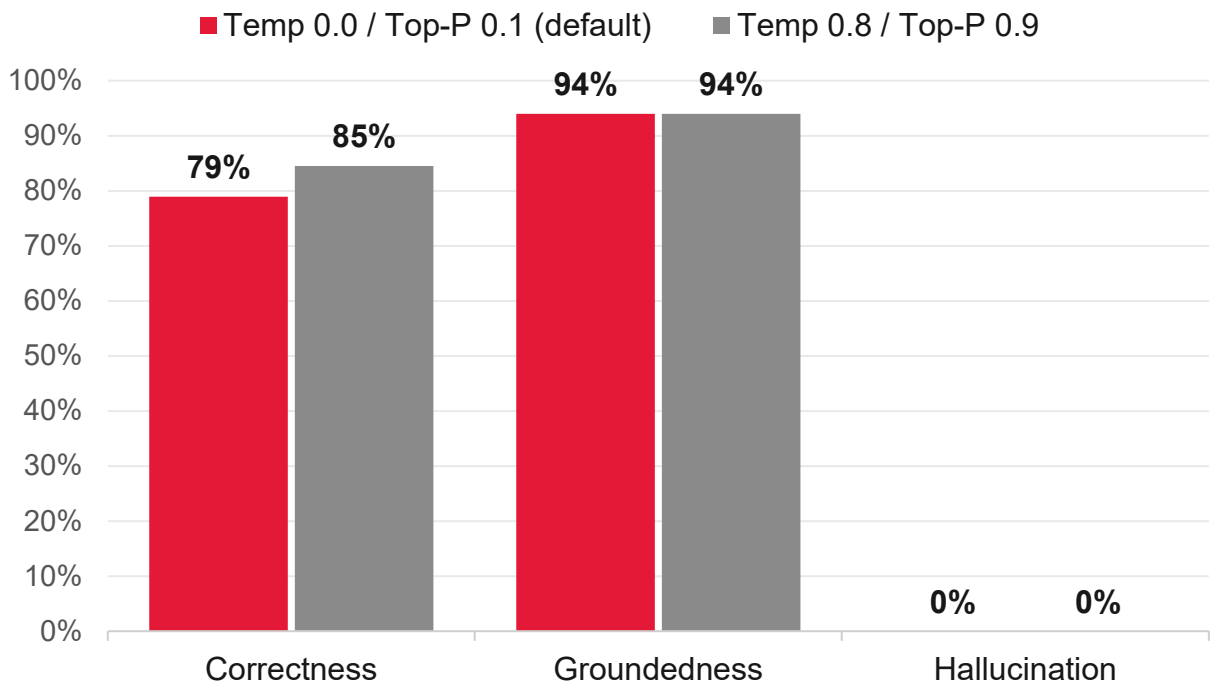
openai/gpt-oss-120b

Why LLM over SLM?

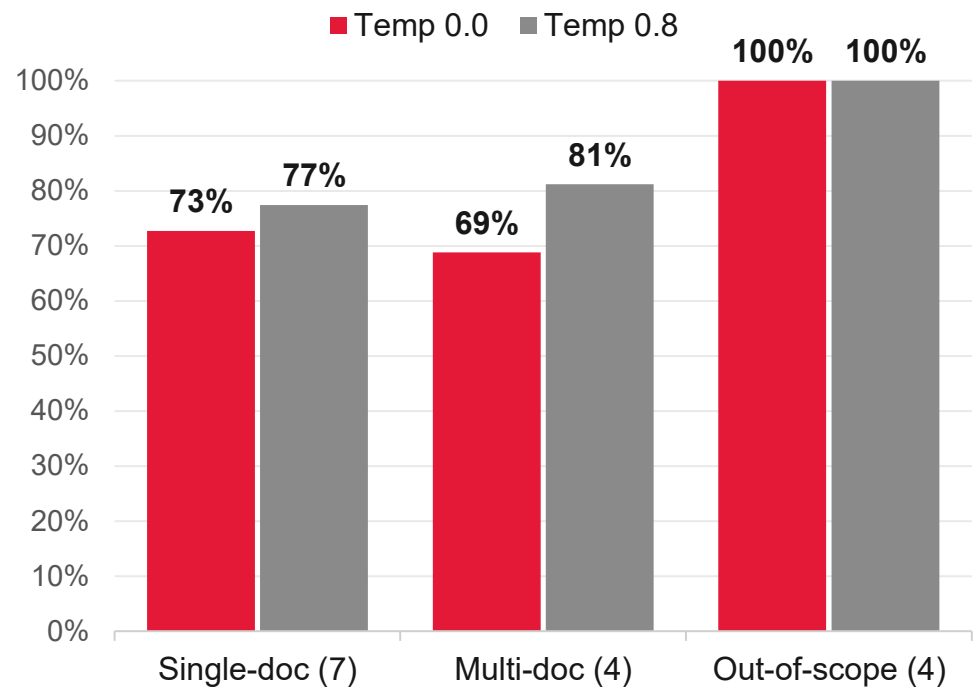
Cross-Policy Reasoning: SLMs miss clauses across overlapping documents; the LLM accurately connects complex multi-policy rules.
Zero Hallucination: SLMs tend to fabricate answers when uncertain; the LLM strictly refuses out-of-scope questions.

No hallucinations flagged across 15 test questions

OVERALL (15 QUESTIONS)



CORRECTNESS BY QUESTION TYPE



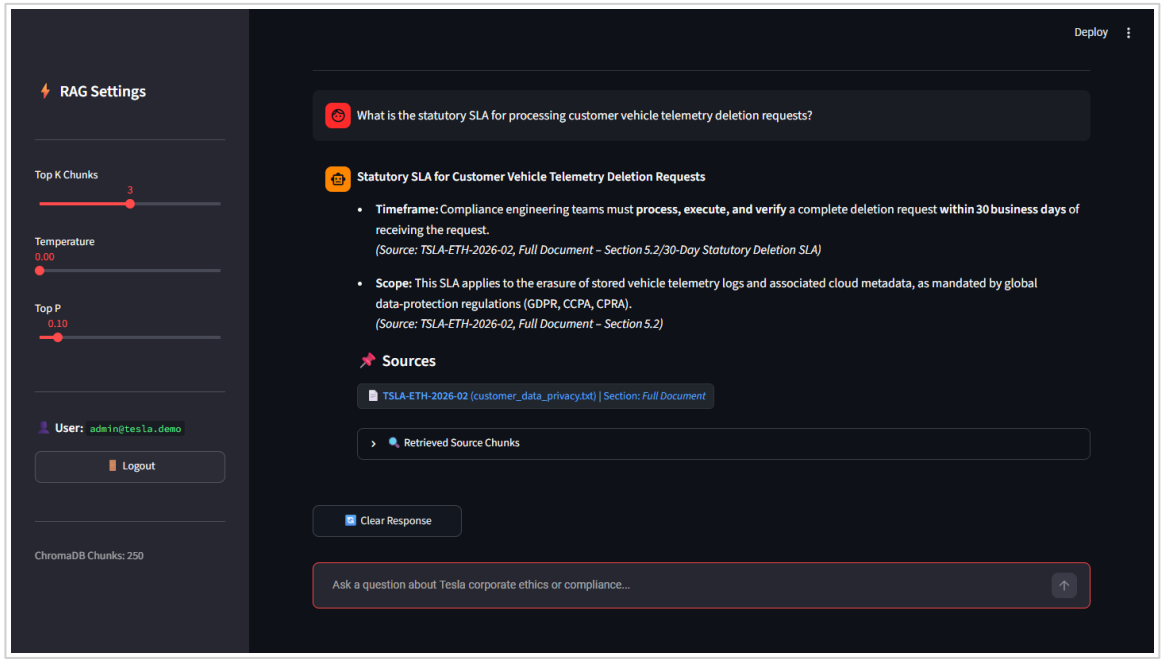
0%
hallucination flagged, both configs

4 / 4
out-of-scope questions declined

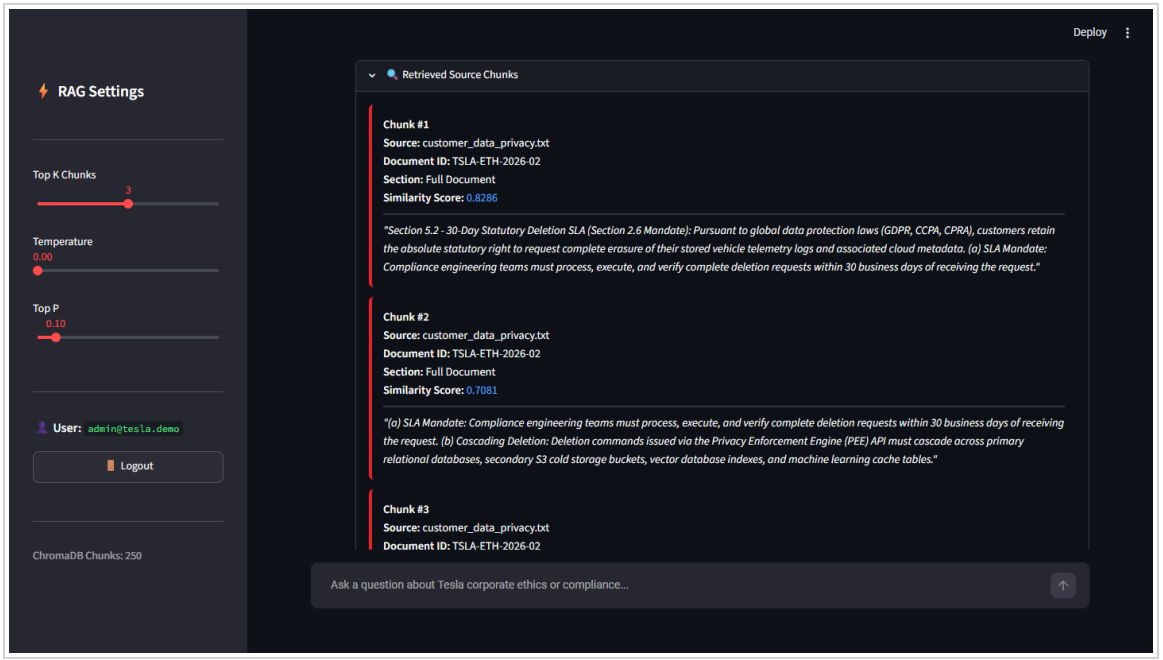
~2s
avg latency per answer (approx.)

No clear gap
temp 0.0 vs 0.8, 15 questions

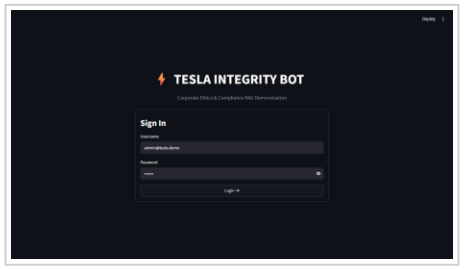
The working application



1 **Grounded answer with source badge**



2 **Retrieved chunks with similarity scores**



3 **Mock sign-in**

Unaltered captures of the running Streamlit app on the synthetic corpus. Any username is accepted at sign-in (mock gate). Sidebar exposes Top-K, temperature and top-p.

Thank You